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• Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

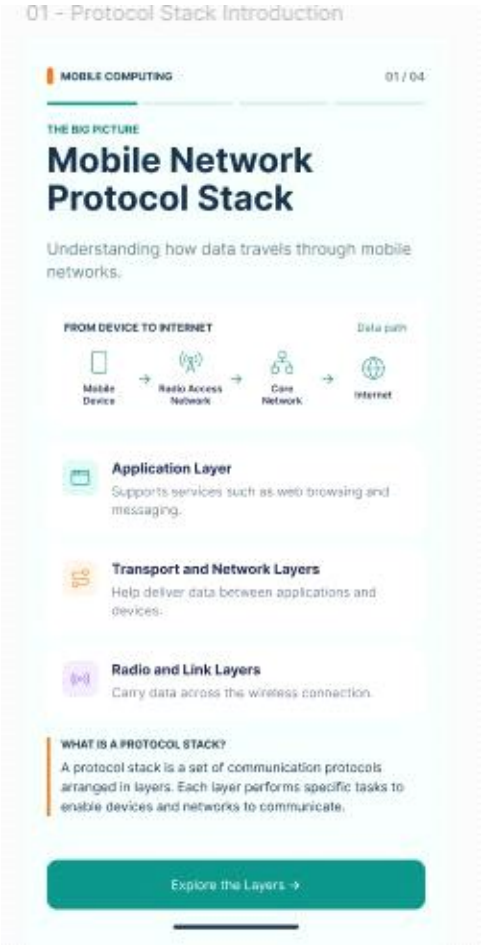
Aim

To create a visual representation in Figma highlighting the key elements of mobile network protocol stacks and understanding the functions of different protocol layers.

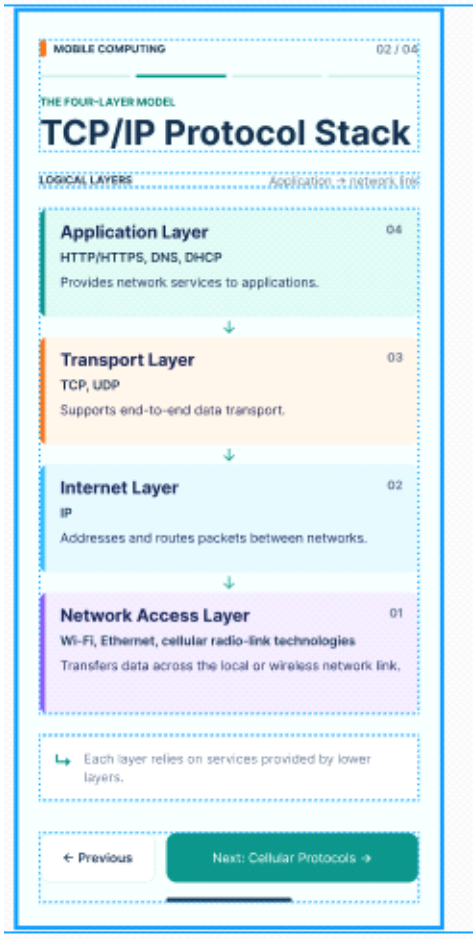
Procedure

- Open Figma and create a new design file.
- Create a single page with the title “Key Elements of Mobile Network Protocol Stacks.”
- Draw five horizontal rectangular blocks representing Application, Transport, Network, Data Link, and Physical layers.
- Add protocols and functions to each layer: HTTP/HTTPS, DNS, DHCP; TCP, UDP; IP, ICMP; Wi-Fi MAC, Ethernet; and Wi-Fi radio and cellular radio interface.
- Use different colors to distinguish the layers and add arrows to show transmission and reception.
- Create a simple mobile network diagram showing Mobile Device → Wireless Radio Link → Base Station → Core Network → Internet/Server.
- Add a reference section describing LTE, 5G NR, GSM, and the functions of PHY, MAC, RLC, PDCP, RRC, and NAS.
- Add a data-flow summary and legend. Align all elements and check the diagram for readability and technical accuracy.

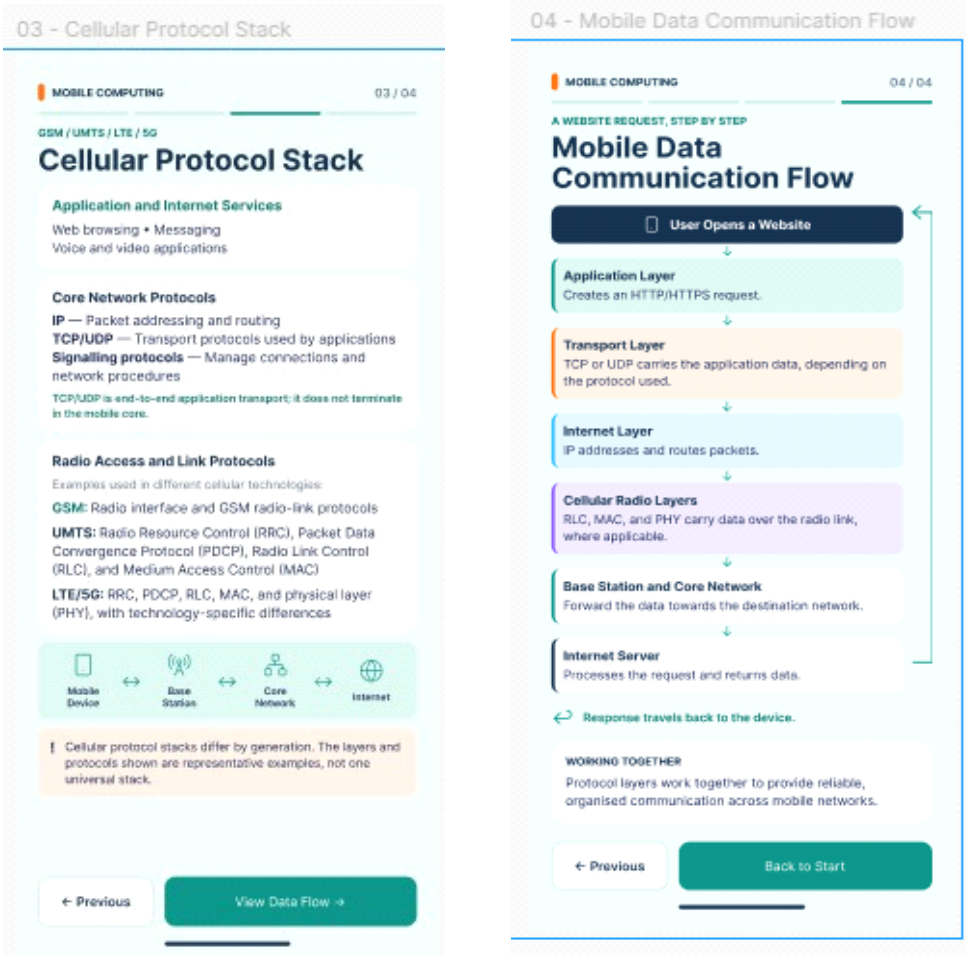
01 - Protocol Stack Introduction



02 - TCP-IP Protocol Stack




01 - Protocol Stack Intr...
Mobile Network Protocol Stack
02 - TCP-IP Protocol St...
TCP/IP Protocol Stack
Application Layer
Transport Layer
Internet Layer
Network Access Layer
03 - Cellular Protocol St...
Cellular Protocol Stack
04 - Mobile Data Comm...
Mobile Data Communication Flow


03 - Cellular Protocol Stack
MOBILE COMPUTING
03 / 04
GSM / UMTS / LTE / 5G
Cellular Protocol Stack
Application and Internet Services
Web browsing • Messaging
Voice and video applications
Core Network Protocols
IP — Packet addressing and routing
TCP/UDP — Transport protocols used by applications
Signalling protocols — Manage connections and network procedures
TCP/UDP is end-to-end application transport; it does not terminate in the mobile core.
Radio Access and Link Protocols
Examples used in different cellular technologies:
GSM: Radio interface and GSM radio-link protocols
UMTS: Radio Resource Control (RRC), Packet Data Convergence Protocol (PDCP), Radio Link Control (RLC), and Medium Access Control (MAC)
LTE/5G: RRC, PDCP, RLC, MAC, and physical layer (PHY), with technology-specific differences
Mobile Device
Base Station
Core Network
Internet
Cellular protocol stacks differ by generation. The layers and protocols shown are representative examples, not one universal stack.
← Previous
View Data Flow →
04 - Mobile Data Communication Flow
MOBILE COMPUTING
04 / 04
A WEBSITE REQUEST, STEP BY STEP
Mobile Data Communication Flow
User Opens a Website
Application Layer
Creates an HTTP/HTTPS request.
Transport Layer
TCP or UDP carries the application data, depending on the protocol used.
Internet Layer
IP addresses and routes packets.
Cellular Radio Layers
RLC, MAC, and PHY carry data over the radio link, where applicable.
Base Station and Core Network
Forward the data towards the destination network.
Internet Server
Processes the request and returns data.
Response travels back to the device.
WORKING TOGETHER
Protocol layers work together to provide reliable, organised communication across mobile networks.
← Previous
Back to Start

Result

A visual representation of mobile network protocol stacks was created in Figma, illustrating the five protocol layers, their functions, data-flow directions, and the relationship between Internet protocols and cellular network protocols.